

# DAG Workflow Engine - ML Pipeline Orchestrator

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## Description

DAG-based workflow orchestration for ML pipelines. YAML task dependencies, topological execution with parallelism, retry with backoff, state persistence.

## Tech Stack

networkx, pydantic, pyyaml, click, rich, pytest

## Features

- YAML DAG configuration
- Topological execution
- Built-in ML tasks
- Retry with backoff
- State persistence
- Rich terminal UI

## Sample Input

YAML pipeline

```
dagflow run --pipeline ml_pipeline.yml
```

## Sample Output

Pipeline execution

Complete: 6/6 tasks passed (12.3s, 2 parallel generations)

## Project Files

```
.gitignore
project-idea.json
pyproject.toml
README.md
.github/workflows/test.yml
dag_workflow_engine/cli.py
dag_workflow_engine/dag.py
dag_workflow_engine/executor.py
dag_workflow_engine/loader.py
dag_workflow_engine/models.py
```

```
dag_workflow_engine/state.py
dag_workflow_engine/ui.py
dag_workflow_engine/__init__.py
dag_workflow_engine/tasks/builtin.py
dag_workflow_engine/tasks/registry.py
dag_workflow_engine/tasks/__init__.py
dag_workflow_engine.egg-info/dependency_links.txt
dag_workflow_engine.egg-info/entry_points.txt
dag_workflow_engine.egg-info/PKG-INFO
dag_workflow_engine.egg-info/requires.txt
dag_workflow_engine.egg-info/SOURCES.txt
dag_workflow_engine.egg-info/top_level.txt
examples/ml_pipeline.yml
examples/parallel_pipeline.yml
examples/simple_pipeline.yml
tests/test_dag.py
tests/test_executor.py
tests/test_integration.py
tests/test_loader.py
tests/test_models.py
tests/test_state.py
tests/test_tasks.py
tests/__init__.py
```

## Test Results

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**PASSED**

Run: 47 | Passed: 47 | Failed: 0